

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-0959

Arkoma 138 kV
January 19, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0959. The Interconnection Customer proposes a 57.4 MW solar photovoltaic facility interconnecting at the Arkoma 138 kV substation in the Pennsylvania area. The request was received on March 05, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 292 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 18 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$21,991,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Havelock / Waukena 138kV	Lighthouse Hill / Stewart Mtn Dam 345kV	164.7	178	33.2
Belfield / Gunter'sville Dam 138kV	Sugar Hill / Ritter Park 345kV	163.6	181	5.5
Belfield Substation / Marquis 138kV Line	Ritter Park / Sugar Hill 345kV	155.4	200	5.5
Stewart Mountain Dam / Shepaug 138kV	Ritter Park / Sugar Hill 345kV	151.9	339	9.2
Waukena / Arkoma Substation 138kV L	Stewart Mountain Dam / Lighthouse Hill 345kV Line	150.2	203	33.2
North / Shepaug 138kV	Ritter Park / Sugar Hill 345kV	150.2	336	9.2
Friendly Hills / Marquis 138kV	Ritter Park Substation / Sugar Hill Substation 345kV	149.4	217	5.5
Friendly Hills / Uc Davis 138kV L	Ritter Park / Sugar Hill 345kV	147.1	231	5.5
Ritter Park / Uc Davis 138kV	Ritterpark / Sugar Hill 345kV	144.9	245	5.5
Havelock / Blountsville 138kV Line	Lighthouse Hill / Stewart Mtn Dam 345kV line	141.3	196	33.2
Williams / Lewisville 138kV	Ritter Park / Sugar Hill 345kV	125.3	253	5.2
Gunter'sville Dam / Sugar Hill Substation 138kV line	Ritter Park / Sugar Hill Substation 345kV	124.7	217	5.5
Torrey / Tostn 138kV L	Stewart Mountain Dam / Ritter Park 345kV	114.5	224	16.6
Blountsville / Vista 138kV L	Stewart Ountain Dam 345-138 kV Autotransformer	107.3	208	12.3

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Cuyamungue Tap Jmec / Lewisville 138kV	N / Holland Park 138kV	106.9	168	6.2
Bullard / Toston 138kV	Stewart Mtn Dam / Ritter Park 345kV	103.4	235	16.6
Newbury Substation / Crescent 138kV	Stewart Mountain Dam / Lighthouse Hill Substation 345kV	101.9	204	13.5
Torrey / Blountsville 138kV	Stewart Mtn Dam / Ritter Park 345kV line	101.7	264	16.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Waukena / Havelock 138kV	Rebuild line with higher-capacity conductor	\$5,194,000	100% (\$5,194,000)
Belfield / Gntersville Dam 138kV	Rebuild line with higher-capacity conductor	\$2,163,000	4% (\$93,000)
Belfield Substation / Marquis 138kV Line	Rebuild line with higher-capacity conductor	\$3,323,000	4% (\$143,000)
Stewart Mountain Dam / Shepaug 138kV	Rebuild line with higher-capacity conductor	\$2,228,000	100% (\$2,228,000)
Waukena / Arkoma Substation 138kV L	Rebuild line with higher-capacity conductor	\$1,830,000	100% (\$1,830,000)
North / Shepaug 138kV	Rebuild line with higher-capacity conductor	\$3,860,000	100% (\$3,860,000)
Friendly Hills / Marquis 138kV	Rebuild line with higher-capacity conductor	\$1,504,000	4% (\$65,000)
Friendly Hills / Uc Davis 138kV L	Rebuild line with higher-capacity conductor	\$3,149,000	4% (\$136,000)
Ritter Park / Uc Davis 138kV	Rebuild line with higher-capacity conductor	\$2,765,000	4% (\$119,000)
Havelock / Blountsville 138kV Line	Rebuild line with higher-capacity conductor	\$2,665,000	100% (\$2,665,000)
Lewisville / Williams 138kV line	Rebuild line with higher-capacity conductor	\$2,021,000	11% (\$222,000)
Guntersville Dam / Sugar Hill Substation 138kV line	Rebuild line with higher-capacity conductor	\$2,207,000	4% (\$95,000)
Torrey / Tostn 138kV L	Rebuild line with higher-capacity conductor	\$2,541,000	25% (\$638,000)
Blountsville Substation / Vista 138kV L	Reconductor line	\$642,000	100% (\$642,000)
Cuyamungue Tap Jmec / Lewisville Substation 138kV Line	Reconductor line	\$3,045,000	100% (\$3,045,000)
Toston / Bullard 138kV L	Reconductor line	\$961,000	25% (\$241,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Crescent / Newbury 138kV	Reconductor line	\$545,000	100% (\$545,000)
Torrey / Blountsville 138kV	Reconductor line	\$916,000	25% (\$230,000)
	Total allocated to project		\$21,991,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.